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ARTICLE

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Leveraging flow mechanics to determine critical process and scaling parameters in a continuous viral inactivation reactor

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Abstract

A continuous viral inactivation (CVI) chamber has been designed to operate with acceptable residence time distribution (RTD) characteristics. However, altering the CVI's geometry and operation to accommodate the scale was not obvious. In this work, we elucidate the influence of Dean vortices and leverage the transition into the weak turbulent regime to establish relationships between input variables and process outputs. This study was targeted to understand and quantify the impact of viscosity, Dean number, internal diameter, and path length on the RTD. When the Dean number exceeds 70, radial mixing generated by the Dean vortices began to consistently alter the axial dispersive effects experienced by the pulse injection. Increasing to a Dean number of >100, the axial dispersive effects were dominated by the Dean vortices which allowed the calculation of the minimum and maximum residence time to be generated. This work provides a method to calculate operational solutions for a tubular incubation reactor in terms of path length, internal diameter, flow rate, and target minimum and maximum residence time specifications that assures both viral residence times while also establishing criteria to maximize product quality during continuous operation.

KEYWORDS

continuous manufacturing, Dean vortices, scale down, scale up, viral inactivation

1 | INTRODUCTION

Low pH viral inactivation is a common step in the manufacturing of therapeutic monoclonal antibodies (mAb) in which the post Protein A elution pool is acidified to a predefined pH and incubated for a predetermined time. When operated correctly, this step is highly robust and effective at inactivating enveloped viruses (ASTM, 2012; Brorson et al., 2003; Chinniah, Hinckley, & Connell-Crowley, 2016). However, in the evolution of biopharmaceutical manufacturing, the pursuit of creating a fully continuous process requires modification of this step (Johnson, Brown, Lute, & Brorson, 2017). This work aims to enlighten the continuous viral inactivation (CVI) methodology where acid is continuously added to the product stream exiting a unit operation and subsequently held in a plug flow reactor (PFR) for a defined residence time (Fiadeiro, Kublbleck, Fahrner, & Salm, 2016; Xenopoulos, 2017).

It is a well-published observation that controllable process set points for a PFR, such as flow rate and reactor volume (Levenspiel, 1999), have a profound influence on product residence time distribution (RTD). However, converting the static batch unit operation, a cornerstone of the viral mitigation strategy, to a fully continuous operation is not trivial. Multiple publications have incrementally strengthened the argument for a robust CVI by addressing theoretical considerations (Johnson et al., 2017; Klutz, Lobedann, Bramsiepe, & Schembecker, 2016), modeling flow characteristics through computational fluid dynamics (Parker et al., 2018), pulse injecting tracers in water for multiple CVI reactor designs (Orozco et al., 2017), as well as establishing that the CVI concept can in fact inactivate virus (Gillespie et al., 2019). This study differentiates itself by examining different scales and addressing the impact of the viscosity gradient, which is expected from a Protein A elution peak, on the minimum residence time (T_{min}) and the maximum residence time (T_{max}) of the product stream. With the removal of the pooling step after the Protein A step, the mAb exits the column in the form of a peak resembling an asymmetric Gaussian curve. This results in protein concentration changing as a function of time as it enters the CVI. The feed stream's viscosity is heavily dependent on protein concentration (Li et al., 2014), therefore viscosity changes as a function of time as well. To move forward with developing this methodology, the influence of this variable protein concentration on viral safety must be addressed.

This study also addresses the scale-up/down methodology. Preliminary testing of these reactors revealed that simply matching the average residence time between scales was not sufficient to match the RTD of the pulse injections. The functional operation of the CVI not only relies on proper definition of the product incubation but also the concept of pressure drop. As the entire continuous operation is scaled up, volumetric flow rates of the process increase as well. Therefore, a single fixed CVI reactor internal diameter that is increased in length to meet residence time requirements is not a viable solution. The definition of the required CVI scale can be condensed down to the productivity of the feeding bioreactor, Protein A column sizing, and required flow rates in the integrated system. The effect of solution viscosity on pressure drop is considered negligible relative to the influence of path length, internal diameter, and flow rate. This study outlines a methodology to identify the CVI reactor specifications to meet product and process requirements.

The CVI design for this work is the "Jig in a Box" (JIB) design (Parker et al., 2018), which is characterized by a repeating serpentine-like pattern. This design utilizes Dean vortices to enhance radial mixing to maximize the efficiency of the reactor. Dean vortices are characterized by the Dean number, which is a ratio of the inertial and centripetal forces to the viscous forces. The Dean number is defined by Equation (1), where ρ is the fluid density, u is average linear flow velocity, D is flow path internal diameter, μ is dynamic viscosity, and R_c is the radius of curvature of the serpentine pattern. Since viscosity is an influencing characteristic of Dean number, it therefore should impact the efficacy of the reactor.

$$\mathrm{De} = \left(\frac{\rho u D}{\mu}\right) * \sqrt{\frac{D}{2R_c}}. \quad (1)$$

Currently, the impact of viscosity and Dean number on the RTD remains uncertain. Previous work from this group concluded that the viscosity parameter had failed to influence the RTD when testing variable viscosity pulse injections (<2.5% v/v of the reactor volume) under a specific range of Dean number values when chased by water (Amarikwa, Orozco, Brown, & Coffman, 2018). In practice, the volume of the protein peak can exceed the volume of the reactor. This study takes the approach of making the mobile phase and pulse injection isocratic in terms of viscosity, density, and expands the window of Dean number values previously studied to inform real-world application and operation.

2 | MATERIALS AND METHODS

2.1 | Materials

The JIB was designed from previous development projects (Parker et al., 2018; Orozco et al., 2017), and was 3D printed utilizing SLA Technology by 3D Systems (Rock Hill, SC). The riboflavin and dextrose used in creating the mobile phases and pulse tracer were purchased through Thermo Fisher Scientific (Suwanee, GA). The viscosities of the solutions were determined by a microVISC Viscometer utilizing an A05 Chip (San Ramon, CA). The densities of the solutions were determined by a Mettler Toledo Densito density meter (Columbus, OH).

2.2 | Residence time distribution generation

The small scale and mid-scale 3D printed JIB were tested using an Akta Avant 150, while the large scale JIB was tested using an Akta Pilot 600 by GE Healthcare (Uppsala, Sweden). The scale of the JIB was defined by the ability to accommodate bench-scale testing (i.e., small scale), 100 L perfusion bioreactor (i.e., mid scale), and >500 L perfusion bioreactor (i.e., large scale). The JIB was first flushed with 1 reactor volume of the mobile phase. Next a fixed volume of riboflavin, 2.5% v/v of reactor volume dissolved in the mobile phase, was pulse injected and chased out with two reactor volumes of the mobile phase. The diffusion of riboflavin during an experiment was considered and deemed not impactful. Specific discussion of diffusion and relevance of riboflavin as a tracer for a PFR has been discussed in previous works (Amarikwa et al., 2018). The injection produced the RTD profiles upon exiting the reactor detected and quantified by UV–vis absorbance at riboflavin's absorbance maximums (Murai, 1991; i.e., 267, 372, and 445 nm).

The internal diameters, number of JIBs connected in series, flow rates, and mobile phases tested in this study are outlined in Table 1. To increase the viscosity of the mobile phase, dextrose was added (i.e., 50, 100, and 200 g/L) to simulate the kinematic viscosity of a mAb solution of ~10, 20, and 50 g/L, respectively. Since CVI is intended to be operated immediately post protein A, peak max of the elution should not reach a concentration high enough to demonstrate non-Newtonian behavior. In addition, the propensity of excipients, which can be non-Newtonian, to be present in the product stream at this step is relatively low. Therefore, the Newtonian fluid of dextrose was considered to be an appropriate model. The RTD peaks were then analyzed by fitting a Gaussian distribution as described in Orozco et al. (2017). A Gaussian distribution maps the RTD for a delta function entering the reactor. During operation, the asymmetric Protein A elution peak can be thought of as of an infinite series of delta pulses. From this fit the variance of the peak, (σ^2_{time}), which is a measurement of the spread of the RTD, was determined. To better understand the influence of the quantitative value of the variance, Equation (2) was calculated, where Q is the volumetric flow rate. Additionally, the data set was converted using Equations (3) and (4) where HETP is height equivalent to a theoretical plate, T_{Ave} is the

TABLE 1 Mobile phases, path length, and flow rate combinations tested via riboflavin pulse injection experiments using the JIB design

Scale	Internal diameter (cm)	Number of JIBs connected in series	Mobile phase	Flow rates tested
Small scale	0.156	1 (L = 11.09 m)	DI Water	1–16 ml/min
Mid scale	0.635	1 (L = 16.44 m)	DI Water	5–100 ml/min
		2	DI Water	
		6	DI Water	
		1	50 g/L Dextrose	
		1	100 g/L Dextrose	
		1	200 g/L Dextrose	10–125 ml/min
Large scale	2.008	1 (L = 16.60 m)	DI Water	25–700 ml/min

mean residence time, RV is reactor volume, and L is the length of the flow path of the JIB.

$$\sigma_{\text{Volume}} = \sqrt{\sigma_{\text{time}\,*}^{2} Q^{2}}, \tag{2}$$

$$\text{HETP} = \frac{\sigma_{\text{time}\,*}^{2} L}{{T_{\text{Ave}}}^{2}}, \tag{3}$$

$$T_{\text{Ave}} = \frac{\text{RV}}{Q}. \tag{4}$$

3 | RESULTS AND DISCUSSION

3.1 | Flow rate and path length (Mid scale)

As seen in Figure 1a, the slowest flow rate produces the widest peak for all three reactor path lengths. This is shown in the graph as having a relatively high standard deviation (e.g., σ_{Volume} ~ 82 ml for the JIB(1) data set at 5 ml/min) which describes the volume distance from the center of the peak (i.e., T_{Ave}) to 34% of total mass to the left or right. As the flow rate increases the peak becomes narrower at an exponential rate (e.g., σ_{Volume}~ 58 ml at 20 ml/min). However, an inflection point is attained when the flow rate increases above 20 ml/min. The peak becomes wider until another inflection point is reached at the 30 ml/min set point. With the flow rate increasing higher, the peaks start to become substantially narrower until the 55 ml/min set point. In total, the progression from slower to faster flow rate displays an initial asymptote, two inflection points, and finally an asymptote. Looking at the JIB(1), JIB(2), and JIB(6) data in Figure 1a, this phenomenon of two asymptotes and two inflection points is maintained at identical flow rates across all path lengths. The only difference is that the standard deviation increases as length increases for a given flow rate, which is a well-characterized observation for PFRs (Levenspiel, 1999). When the data points from Figure 1a are translated using Equation (3) and converted into HETP, Figure 1b is created making the three path length's data sets overlay.

To understand the driving force for the two inflection points, further exploration into previously published work on Dean vortices was undertaken. In Aider et al. (2005), flow patterns were visualized using suspensions in water when between two rotating drums (i.e., Taylor–Couette flow). As the flow increased in velocity in the flow cell, Aider made observations at specific Dean number where the flow patterns shifted from laminar to chaotic. A similar experiment was conducted in the JIB using suspended mica in water and found the same laminar to a chaotic flow transition. These specific Dean numbers and corresponding observations outlined in Aider et al are coplotted with the JIB(1) data points in Figure 2. The two inflection points and the faster flow rate asymptote correspond to the visually observable manifestations of the flow transitioning from the onset of unstable flow, undulated

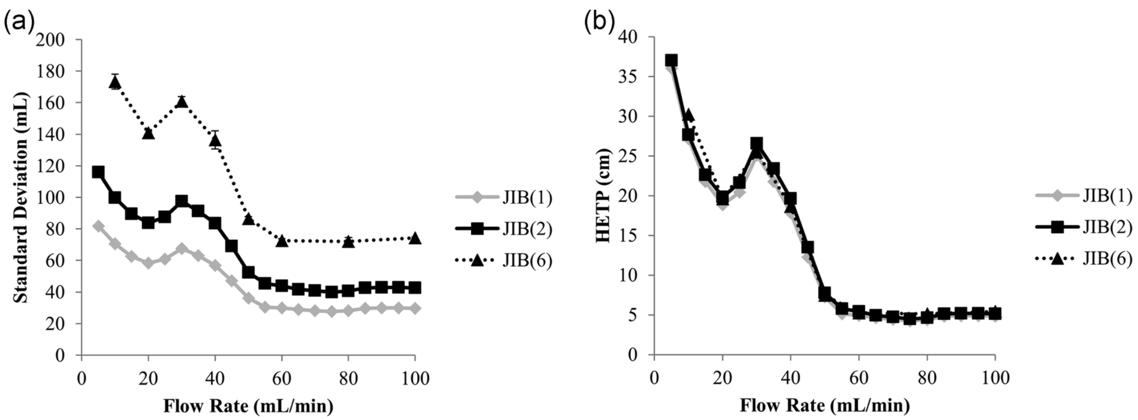

FIGURE 1 Reaction of exiting pulse injections as a function of flow rate and path length (i.e., 1, 2, or 6 JIBs connected in series) of mid-scale Jig in a Box (JIB) operation. (a) The pulse experiences two asymptotes and two inflection points at identical flow rates at different path lengths. Error bars representing (+/−) the sample standard deviation are plotted. Data points with no error bars had a small enough difference that the dot symbol covers the bars. (b) Peak spreading is normalized to height equivalent to a theoretical plate (HETP) along the y-axis by Equation (3), allowing peak spreading to be calculated based off-path length and flow rate

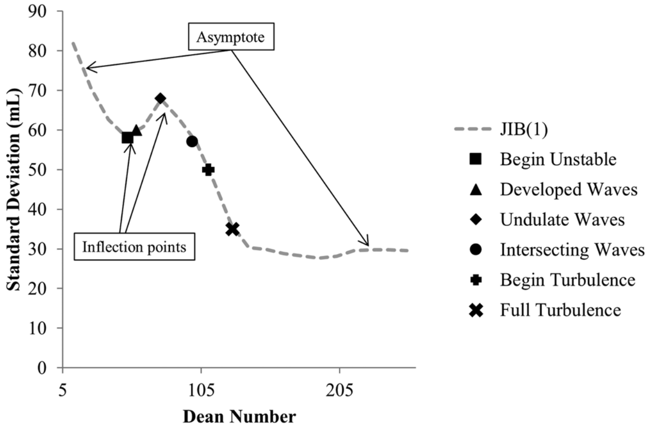

FIGURE 2 Correlation of the Dean number required to generate transitions in Taylor–Couette flow (Aider, Skali & Brancher, 2005) and the two inflection points and asymptotes created in the spreading of the residence time distribution (RTD) profiles of the Jig in a Box (JIB)

waves, and full turbulence, respectively. The onset of turbulence for flow in a circular straight pipe is typically observed at a Reynolds number of ~2,000. The JIB was able to simulate turbulent flow behavior at a Reynolds number of ~174. Due to this large discrepancy, the term "weak turbulence" was used.

3.2 | Changes in viscosity (Mid scale)

To prove the validity of the claim that the shift of two asymptotes and two inflection points behavior found in the section above was controlled by the Dean number, the mobile phase's viscosity was increased with three concentrations of dextrose. Here, dextrose was used as a surrogate for the protein to be able to increase viscosity at different concentrations. Figure 3a displays the reaction of the addition of the dextrose. For all four mobile phases, the standard deviation began at its highest value at the slowest flow rate and became narrower with the increase in flow rate. However, with the increase in viscosity, the flow rate required to reach inflection point 1 increased. The same is true for inflection point 2 and the flow rate required to reach the second asymptote.

To account for this apparent shift in the inflection points and asymptote, the x-axis was normalized by converting the flow rate to Dean number described in Equation (1) and shown in Figure 3b. Once normalized to the Dean number, the asymptotes and inflection points align. While the shift in the data set returns to normal with the correction, the magnitude of the spreading appears only to be corrected for the higher Dean number (i.e., De > 70) operation. This may be due to Dean vortices taking over the radial mass transfer above De = 70 in the form of "undulate waves" as observed by Aider et al. (2005). For the lower Dean numbers (De < 70), other dispersion mechanisms appear to have influence.

Overall, the lower the Dean number, the less chaotic the Dean vortices were. This resulted in the JIB becoming less efficient increasing the HETP. The more laminar the flow behavior was, the higher the HETP value. Since laminar flow is dominated by dispersion, the no-slip boundary condition made the pulse injections broader (higher HETP). As the Dean number increases, radial mixing increases making the pulse injections narrower.

3.3 | Using Dean number to calculate residence time distribution (Mid scale)

To inform the operation of the JIB, a correlation between viscosity, flow rate, and path length can be used to calculate the RTD. This approach exploits the relationship of the RTD profiles and the Dean number. In this case, a criterion that the JIB unit operation must maintain a Dean number of >100 was established. The HETP values were pooled from the various dextrose concentrations and flow rate experiments and only included data points operated at a Dean

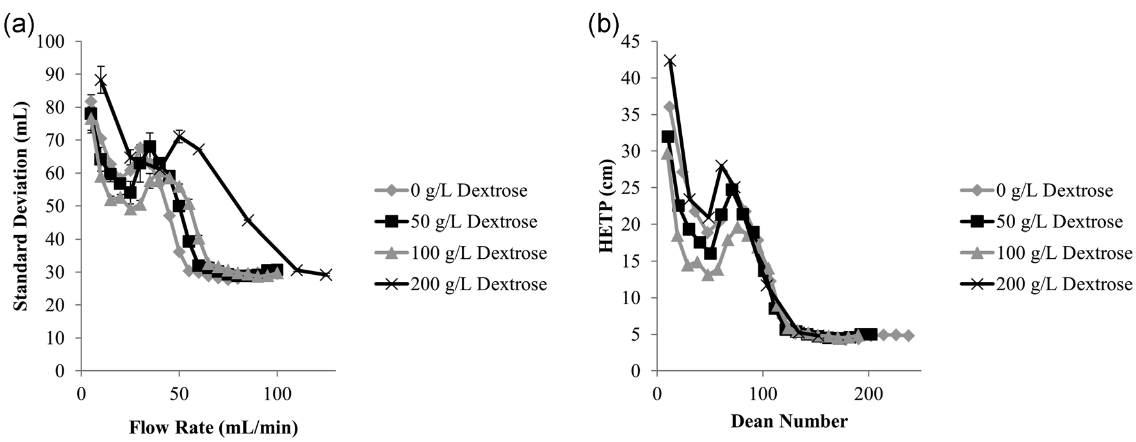

FIGURE 3 Reaction of exiting pulse injections as a function of flow rate and viscosity of mid-scale Jig in a Box (JIB) operation. (a) The pulse experiences two asymptotes and two inflection points, but at different flow rates. Error bars representing (+/−) the sample standard deviation are plotted. Data points with no error bars had a small enough difference that the dot symbol covers the bars. (b) Peak spreading is normalized along the x-axis by Equation (1) and the y-axis by Equation (3), allowing the two asymptotes and two inflection points to align

number >100 (Figure 4a). From this data set, a third order polynomial can be fit to this data pool (Figure 4b) resulting in Equation (5), where a, b, c, and d are based on the polynomial fit. Equation (5) allows for the calculation of HETP as a function of the Dean number. This can be used in combination with Equations (1,3) to use flow rate, viscosity, and path length to determine peak spreading (σ^2_{time}). A parity plot can be generated for this correlation (Figure 4c) which describes how well the polynomial calculates the volumetric spreading of the pulse injection in the JIB across path lengths, flow rates, and viscosities where De > 100. With an R^2 value of 0.9201, the fit is reasonably predictable given the inherent error due to the oscillations of the third order polynomial. When the HETP is converted into minimum residence time or $T_{min}(5\sigma)$, which will be discussed later, the calculation across flow rates, viscosities, and JIB lengths is highly accurate as displayed in Figure 4d as the second parity plot.

$$\mathrm{HETP} = a\mathrm{De}^3 + b\mathrm{De}^2 + c\mathrm{De} + d. \tag{5}$$

3.4 | Leveraging Dean number to advise scale-up/down methodology

One way to scale the JIB up or down is to alter the internal diameter of the flow path. This will be referred to as “vertical scaling.” The JIB design was altered such that the internal diameter was increased to 2 cm (i.e., large scale) and decreased to 0.156 cm (i.e., small scale) while also maintaining the aspect ratio (i.e., the ratio between the radius of the flow path and the radius of curvature of the serpentine path). Comparing the experimental data series outlined in Table 1 and displayed in Figure 5a, the three scales all display the same shift at similar Dean number values. To make the comparison of operation across scales, the HETP was normalized by the internal diameter of the flow path to create a reduced HETP (Equation (6)), similar to a common methodology in column chromatography.

$$\frac{\mathrm{HETP}}{D} = h. \tag{6}$$

The overlay of the data set (Figure 5b) matches fairly well with the exception of the high Dean number values for the small scale and the low Dean numbers of the large scale. This, however, is possibly due to the equipment. Being that this data set is representitive of dispersion, any flow path the pulse injection travels through before the JIB and before the detector will influence the experimental results. The small scale JIB is affected more by the extra volume of the system than the large scale JIB's which may result in the larger reduced HETP. In the case of the small scale JIB, the highest flow rate data point and the variance of a pulse through column by-pass had similar $\sigma^2_{time.}$

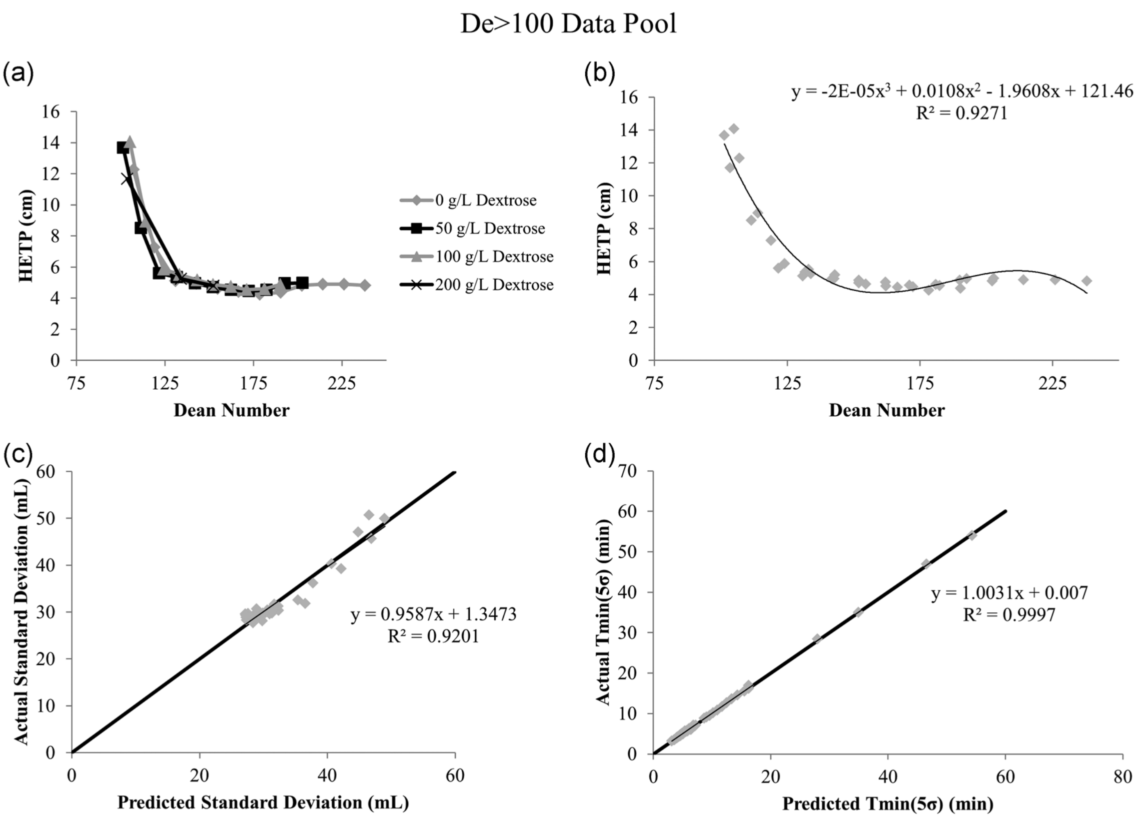

FIGURE 4 (a) Differentiated data points for the four viscosity data pools across flow rates for the mid-scale Jig in a Box (JIB) excluding data point De < 100 (b) Third order polynomial fit to the data set. (c,d) are the resulting parity plots for standard deviation and $T_{min}(5\sigma)$, respectively comparing the experimental and the calculated values from the polynomial fit for path length, flow rate, and viscosity experiments where De > 100

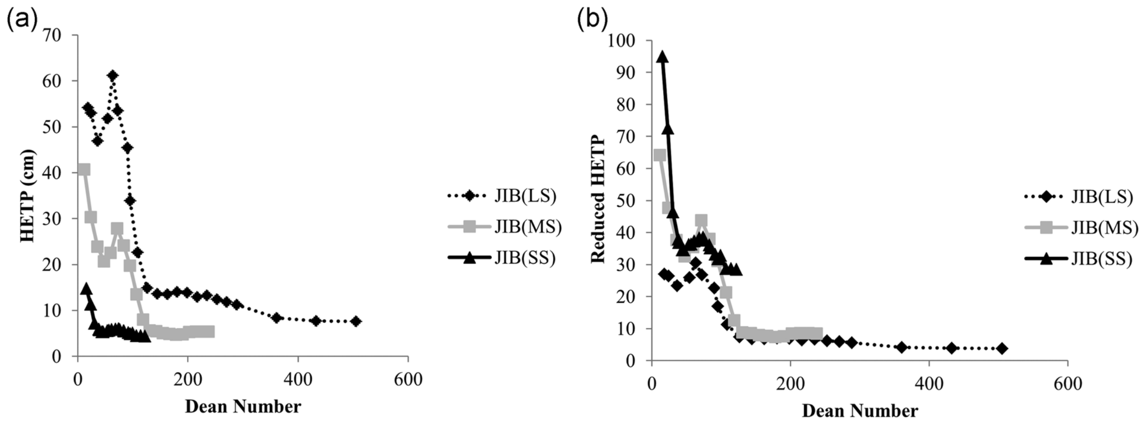

FIGURE 5 Reaction of exiting pulse injections normalized along the x-axis by Equation (1) and the y-axis by Equation (3) from small, mid, and large scale Jig in a Box (JIB) experiments. (a) Similar to previous data sets. Pulse experiences two asymptotes and two inflection points, at similar Dean numbers. (b) Height equivalent to a theoretical plate (HETP) is normalized to reduced HETP by dividing by internal diameter allowing the data sets to nearly overlay

3.5 | Application to unit operation

To implement this unit operation, defining flow rate, path length, internal diameter, maximum viscosity, and incubation requirements is paramount. Given a desired JIB design for a fixed internal diameter, a way to scale this JIB up or down is to alter how many JIB's are connected in series and the operational flow rate. This will be referred to as "horizontal scaling." Applying this methodology, there are two main approaches to design and operate the JIB based CVI unit operation:

1. Given product stream requirements (i.e., $T_{\min}$ and $T_{\max}$), calculate the length of reactor and operational flow rates.
2. Given the size of the reactor and operational flow rate, calculate the product stream $T_{\min}$ and $T_{\max}$.

Starting with a target $T_{\min}$ (i.e., the minimum time the target molecule can be in the acidic condition before impacting viral safety) and a $T_{\max}$ (i.e., the maximum time the target molecule can be in the acidic condition before impacting product quality), Equations (7,8) can be applied to help determine the flow rate and path length required to meet those specifications.

$$T_{\mathrm{Ave}} = T_{\min} + (n * \sigma_{\mathrm{time}}), \tag{7}$$

$$T_{\mathrm{Ave}} = T_{\max} - (m * \sigma_{\mathrm{time}}), \tag{8}$$

$$T_{\mathrm{Taylor}} = \frac{T_{\mathrm{Ave}}}{2}. \tag{9}$$

Table 2 and Figure 6 outline the quantitative aspect of the choice for the "n" and "m" value for σ_{time}. The theoretical onset of the breakthrough of virus and target product starts with T_{Taylor} defined by Equation (9). This value is derived from Hagen and Poiseuille which states that for flow in a circular pipe, the fastest portion of the flow occurs in the geometric center of the cross-sectional area of the flow path and operates at two times the average velocity of flow.

The exiting pulse injection is well approximated by a Gaussian peak with variance σ^2_{time}. For example, $n = 5$ (i.e., $T_{\min}$ (5σ)) is understood to represent that 0.00003% of the product exited the reactor in-between T_{Taylor} and $T_{\min}$ (5σ). This difference between T_{Taylor} and $T_{\min}$(5σ) can be visualized in Figure 6. Table 2 and Figure 6 also highlight $T_{\max}$ decisions. For example, $m = 3$ (i.e., $T_{\max}$(3σ)) would be representative of ~99.865% of the product pool exiting the reactor in less than or equal to $T_{\max}$. This parameter is important since the low pH condition, that is, used to inactivate enveloped viruses also can lead to product degradation affecting product quality of the target molecule. Therefore, $T_{\max}$ decisions would be reliant on the product stability data or accepted yield loss in the presence of the acidic or any other viral inactivating condition. Combining Equations (7,8), we get Equation (10) which allows σ_{time} to be calculated and subsequently T_{Ave} as well using Equation (7,8).

$$T_{\max} - T_{\min} = \Delta T = (n + m) * \sigma_{\mathrm{time}}. \tag{10}$$

Using Equations (3,4), and (5), a locus of path lengths and flow rates can be found (i.e., starting with flow rates that yield a De > 100) can be found to meet specifications resulting in Figure 7a. With

TABLE 2 Tangible quantifications of the standard deviation parameter discussed in this work

Reactor incubation risk assessment		
Time point	Time (min)	Quantity of pulse that has exited the reactor (%)
Viral risk assessment ($T_{\min}$)		
T_{Taylor}	39	0
$T_{\min}$(5σ) ($n = 5$)	61.2	0.00003
$T_{\min}$(3.5σ) ($n = 3.5$)	66.5	0.023
Mid point		
T_{Ave}	78.8	50
Product quality risk assessment ($T_{\max}$)		
$T_{\max}$(2σ) ($m = 2$)	85.3	97.725
$T_{\max}$(3σ) ($m = 3$)	88.7	99.865

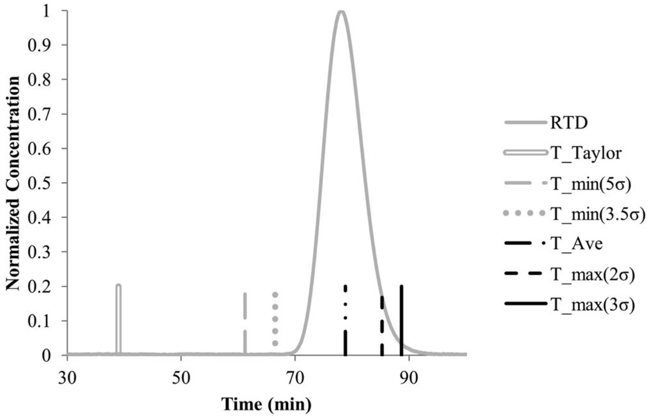

FIGURE 6 Result of a pulse injection experiment in which minimum residence time (T_{min}) and maximum residence time (T_{max}) decisions from Table 2 are coplotted

calculated T_{Ave} and σ_{time} constraints, the minimum flow rate and minimum reactor volume can be calculated. This is an unadvisable location to operate as any increase or decrease in flow rate will result in a violation of either T_{min} or T_{max}. As the reactor volume increases, the volumetric flow rate increases as well to maintain the target T_{max}. When the flow rate increases, the efficiency of the JIB also increases. This allows more freedom in JIB operation displayed as two lines representing the lower and upper limit on Figure 7a. In addition to the increasing efficiency, alteration of the operation window can be modulated by altering the *n*, *m*, T_{min}, or T_{max} values of Equation (6,7).

To visualize this ideology and phenomenon, and given a process where the $T_{min}(5\sigma)$ and $T_{max}(3\sigma)$ are strictly defined at >60 and <79 min, respectively, Figure 7b was generated. For the smaller reactor size operated at a slower flow rate, the σ_{time} value is larger compared to the higher flow rate with a larger reactor size, visualized by the difference in width of the two peaks. Operating at the target flow rates, both designs obey the T_{min} and T_{max} constrains. With the increased sized reactor and corresponding faster flow rate, the σ_{time} decreases allowing flexibility for deviations during operation.

3.6 | Real-world implementation

When the CVI is implemented into real-world operation in a GMP setting, variable flow rates due to pump flow rate variability and viscosity gradients are inevitable and must be accounted for. With the work conducted in this experimental series, this variability can be addressed by understanding process operational extremes and corresponding worst-case conditions and quantify how they impact the residence time of the product stream. In addition, feed stream heterogeneities associated with Protein A elution, such as pH and protein concentration, must also be addressed in terms of reaching the target pH for viral inactivation. However, the topic of inactivation kinetics exceeds the scope of these experiments.

Based on the isocratic viscosity experiments, it appears that the worst case for viral incubation time is a high viscosity solution. Given that a Protein A elution peak's viscosity experiences one peak maximum and will only decrease in concentration as it travels through the reactor through dispersion; the peak max should therefore be considered the worst case. All other portions of the peak (i.e., the front and tail) will have lower viscosities relative to peak max, a larger Dean number, and therefore better radial mixing.

To validate this claim, a mock protein A chromatography elution peak was generated using a binary mixture of dextrose to increase viscosity and NaCl to generate a conductivity trace. In the theoretical case of a protein A column, the mAb will elute from the column as an asymmetric Gaussian curve. This general peak shape was generated using the gradient function of the Akta Avant at a constant flow rate of 50 ml/min through JIB(6) (i.e., six 0.635 cm i.d. JIBs connected in series), where the A1 line contained DI Water, A2 contained riboflavin dissolved in water, and B1 Contained 200 g/L dextrose with ~150 mM NaCl. To evaluate the different viscosity gradient locations, four pulse injection locations were chosen with the first

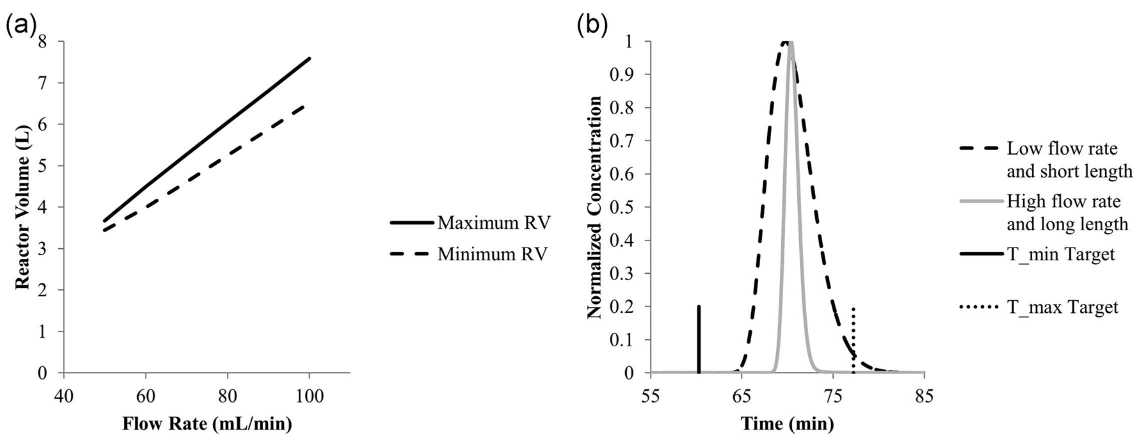

FIGURE 7 (a) Using the correlation generated from Figure 4 to determine the reactor volume (RV) and flow rate combinations to satisfy T_{min} and T_{max} constraints for the mid-scale Jig in a Box (JIB). (b) Resulting residence time distribution (RTD) profiles from two flow rates and reactor size combinations. When operated at their targeted set points, both satisfy the arbitrary but strict requirements of $T_{min}(5\sigma)$ > 60 min and $T_{max}(3\sigma)$ < 79 min

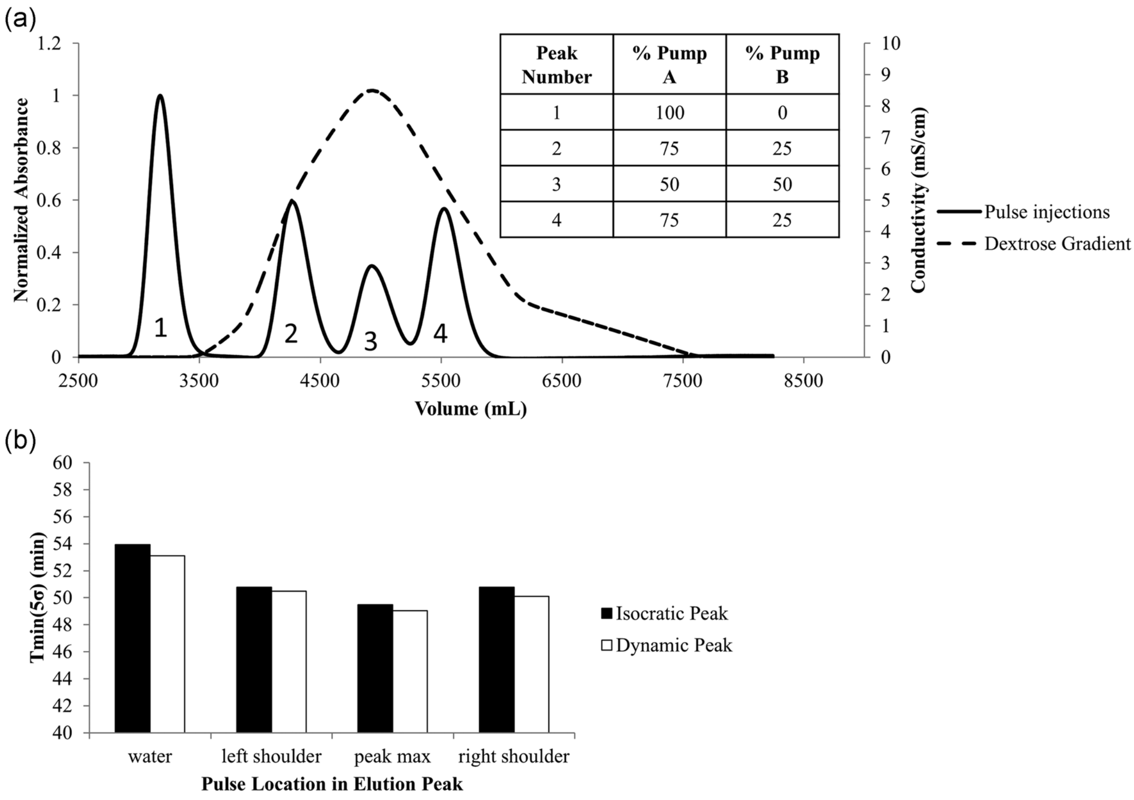

FIGURE 8 (a) Applying pulse injection experiments to a viscosity profile mimicking the dynamic viscosity composition of a protein A elution peak. Injections were applied before the start of the dextrose, the two peak mid-heights (~50 g/L dextrose) and the peak max (~100 g/L dextrose). (b) Comparison of isocratic $T_{min}(5\sigma)$ for 0, 50, and 100 g/L dextrose and corresponding locations at the 0%, 25%, and 50%

occurring before the addition of dextrose, the two peak mid-heights and the peak max yielding dextrose concentrations of 0, 50, 100, and 50 g/L, respectively. Figure 8a displays all four injections. To insert the peak without disturbing the density gradient, pump A was switched from A1 to A2 while maintaining the gradient slope during the riboflavin addition. This explains why the pulse injections are at different heights underneath the dextrose curve. Since the four injections overlapped when conducted in a single experiment, the pulse injection locations were evaluated individually in their own experiment and fitted with a Gaussian distribution and a $T_{min}(5\sigma)$ was calculated.

The phenomena of peak spreading occurred in the dynamic composition setting. Figure 8b shows how well the isocratic data matched the dynamic concentration experiments in terms of $T_{min}(5\sigma)$ mentioned before. The difference between the isocratic and dynamic experimental results was less than 1 min, while the difference between the first water pulse injection and the pulse at peak max was ~4 min verifying it as the worst case for viral incubation.

4 | CONCLUSIONS

The primary objective of this study was to understand the effect of input variables on Dean number and how that propagated to process outputs, specifically T_{min} and T_{max}. During the course of the study, it became clear that the operation of this CVI was reproducible from the input variable standpoint of flow rate, path length, and viscosity. This allowed the fitting of an empirical equation and the corresponding equation-based framework to calculate CVI operation as well as inform scale-up and down methodologies both vertical and horizontal. This study will help inform the operation of the CVI, determine its operation, and help generate a subset of experiments to help to test using bacteriophage and mammalian viruses to prove the validity of the JIB CVI unit operation.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

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